NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

B.TECH/CE/ECE/EE/IT/ME/ODD/1st SEM/R_21//2021-2022
YEAR: 2022

CHEMISTRY-I
CH101

TIME ALLOTTED: 3 HOURS FULL MARKS: 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable

GROUP – A
(Multiple Choice Type Questions)

1. Answer any ***ten*** from the following, choosing the correct alternative of each question: **10×1=10**

SL. NO.		Question	Marks
1.	(i)	In a certain process 678J of heat is absorbed by a system while 290J of work is done on the system. What is the change in the internal energy for the process? a) 970J b) 968J c) 428J d) 972J	1
	(ii)	Ion exchange method is used a) To remove permanent of water b) To protect metal from corrosion c) To measure alkalinity of water d) To measure chloride ion in water	1
	(iii)	IR spectra normally carried on a) Liquid sample b) Gas samples c) Solid samples d) All	1
	(iv)	All naturally occurring processes proceed spontaneously in a direction which leads to a)decrease in entropy b) increase in enthalpy c) increase in free energy d) decrease of free energy	1
	(v)	Magnetic quantum number (m) defines the a) shape of the orbital b) size of the orbital	1

c) orientation of the orbital
d) spin of the electron

(vi) Gas with higher ‘a’ i.e. Vander Waal’s constant of pressure correction 1

a) can be liquefied easily
b) cannot be liquefied easily
c) cannot be predicted
d) no relation

(vii) How many spectral emission lines are conventionally obtained in the emission spectra of H atom 1

a) 3
b) 4
c) 5
d) 2

(viii) Maximum work can be obtained in an isothermal process when it is: 1

a) Irreversible
b) Adiabatic
c) Reversible
d) Cyclic

(ix) Which type of polymer is crystalline? 1

a)Isotactic
b)Syndiotactic
c)Atactic
d)None of these

(x) Example of a natural polymer is: 1

a) PVC
b) Nylon
c) Cellulose
d) Bakelite

(xi) For ideal gases, compressibility factor (Z) is 1

a) >1
b) <1
c) =1
d) 0

(xii) The most stable carbonation among the following is 1

a) CH_3^+
b) $(CH_3)_2CH^+$
C) $(CH_3)_3C^+$
d) $PhCH_2^+$

GROUP – B
(Short Answer Type Questions)
(Answer any three of the following) 3 x 5 = 15

SL. NO.			Marks
2.	(i)	Write down the Schrödinger wave equation in its standard form and define each term in it.	2
	(ii)	2 moles of ammonia are enclosed in a 5 lit flask at 27˚C. Calculate the pressure assuming the gas behaves as a real gas.(a= 4.17 and b= 0.037, a & b has their usual units)	3
3.		What are quantum numbers and what do they signify? How are these numbers related to each other?	5
4.	(i)	IR spectroscopy can efficiently probe H bonding inside the molecule: Explain with a suitable example.	3
	(ii)	Distinguish between inductive effect and electrometric effect.	2
5.	(i)	Why formic acid is stronger than acetic acid?	2
	(ii)	Write vander Walls's equation of state for n moles of a real gas. Explain the terms involved.	3
6.		Write down the chemical structure of repeating units of nylon 6,6. The degree of polymerization is 1000. Find the molecular weight of polyethylene. What is the difference between HDPE and LDPE?	5

GROUP – C
(Long Answer Type Questions)
(Answer any *three* of the following) **3 x 15 = 45**

SL. NO.			Marks
7.	(i)	Write down the limitations of Bohr Postulates.	3
	(ii)	Define electron affinity. How does electro negativity depend on oxidation state and hybridization state of atom?	4
	(iii)	Calculate the ΔG of the reaction occurring in the Daniel Cell $Zn \mid ZnSO_4\,(0.1M) \parallel AgNO_3\,(0.1M) \mid Ag$ at 25 ^{0}C, when E^o_{cell} = 1.56 volt.	3
	(iv)	What are the units of van der Waals' constant a and b?	2
	(v)	What is degree of polymerization of a polymer? Explain with example.	3
8.	(i)	Which drug is more effective – Apirin or Paracetamol? Explain why?	3
	(ii)	Explain the base exchange process of water softening. Why it is called "base exchange process"?	4

	(iii)	Suggest some practicable methods of corrosion control.	4
	(iv)	Explain exothermic and endothermic processes.	3
	(v)	Under what condition $\Delta H = \Delta q$ in terms of Thermodynamics	1
9.	(i)	Write a comparative discussion on SN1 and SN2 reactions involving a) rate equation b) mechanism c) potential energy diagram d) implication of stereochemistry, if any.	4X2=8
	(ii)	Write down the conditions of resonance.	2
	(iii)	Can we liquefy a gas by increasing pressure only? Why?	3
	(iv)	The ionization energy of Na^+ is more than Ne though both of them have same electronic configuration. Explain	2
10.	(i)	What is tacticity? Classify the polymer based on its tacticity.	4
	(ii)	What is hard water? What are the types of hardness? How it can be removed?	5
	(iii)	What is the common reagent in preparation of Paracetamol and Aspirin?	1
	(iv)	Why corrosion is a spontaneous process? Anodic protection is a better method than cathodic protection. Explain.	2
	(v)	Draw and level all possible modes of vibration for CO_2 molecule and indicate the IR active and inactive modes.	3
11.	(i)	Entropy reaches to maximum value at the point of equilibrium. Explain.	2
	(ii)	What is chromophore and auxochrome? Give examples.	3
	(iii)	Highlight the significance of Gibbs equation.	2
	(iv)	$SnCl_4$ exists but Pb forms $PbCl_2$ though both Sn and Pb belong to same group (Group- XIV). Explain with inert pair effect.	3
	(v)	Compare between Thermo-setting and Thermoplastic polymers with examples	5